

VULCAN WERKE

Problem Statement and Solution Design

Vulcan Werke struggles with fragmented data spread across multiple systems that do not communicate with each other. Critical production, quality, and operational information is difficult to access, and much of the knowledge about where data is stored depends on a single employee.

As a result, production issues and quality deviations are identified too late, rework costs increase, and reporting for management, customers, and certifications requires significant manual effort. The lack of consistent and reliable data also makes it difficult to investigate root causes, prove improvements, and make informed operational decisions.

Our solution is to create a centralized data platform that integrates information from all relevant systems into a single, reliable source of truth. By connecting production, quality, and operational data, Vulcan Works gains real-time visibility into performance, faster identification of issues and root causes, and automated reporting capabilities. This reduces manual effort, lowers rework costs, and enables data-driven decision-making across the organization.

Persona	Discovered pain point	How it shapes the solution
Mira Production Manager	– Data from 5 systems arrives in different formats. Mira assembles it by hand before every report. – Outages and defects are found after the line has already stopped — fixing them costs more time and material than catching them early. – No live view of line performance. Crashes are reactive.	– Real-time dashboard per line; alerts fire within seconds of a threshold breach. – Shadow mode runs alongside production before go-live — alerts are tested without interrupting the floor. – UI designed for fast response: alert, reason, action in 3 clicks. – Pilot on 1 line first to show measurable ROI before expanding.
Markus IT / Shopfloor Tech	– 5 source databases, 3 on a separate DEV network with incompatible schemas. Integration is manual and sits entirely with Markus. – Any external data transfer raises GDPR and NDA concerns. Raw machine	– Read-only edge adapters per source — no write access, no schema changes to existing systems. – Raw telemetry stays on-prem. Only aggregated signals go to the central store. – Markus gets a connector management UI so he can add,

	data can't leave the building. – Markus has refused cloud-based tools before. On-prem, read-only is a hard requirement.	pause, and audit sources himself. – Zero changes to existing production systems during rollout.
Ellen / Ingrid Compliance & QM	– ISO 9001 audits require clean, timestamped records. Current data is hard to trace back to source. – Uncertain whether AI adds compliance risk rather than reducing it. – Log files and work-time data fall under employee protection — retention periods and masking are open questions. – Every flagged incident needs a paper trail with a named human decision before escalation.	– Immutable audit snapshots stored locally; records are redacted at ingestion where employee data is involved. – GDPR/DPIA and EU AI Act position documented in plain language before sign-off. – Manual sign-off required on every AI-flagged incident — no automated escalation. – Export templates match ISO 9001 field requirements so reports don't need reformatting.

Solution Design Sketch



